Experiment No:10

Python for Data 26: ANOVA

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In lesson 24 we introduced the t-test for checking whether the means of two groups differ. The t-test works well when dealing with two groups, but sometimes we want to compare more than two groups at the same time. For example, if we wanted to test whether voter age differs based on some categorical variable like race, we have to compare the means of each level or group the variable. We could carry out a separate t-test for each pair of groups, but when you conduct many tests you increase the chances of false positives. The analysis of variance or ANOVA is a statistical inference test that lets you compare multiple groups at the same time.

One-Way ANOVA

The one-way ANOVA tests whether the mean of some numeric variable differs across the levels of one categorical variable. It essentially answers the question: do any of the group means differ from one another? We won't get into the details of carrying out an ANOVA by hand as it involves more calculations than the t-test, but the process is similar: you go through several calculations to arrive at a test statistic and then you compare the test statistic to a critical value based on a probability distribution. In the case of the ANOVA, you use the "f-distribution".

The scipy library has a function for carrying out one-way ANOVA tests called scipy.stats.f_oneway(). Let's generate some fake voter age and demographic data and use the ANOVA to compare average ages across the groups:

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import scipy.stats as stats
```

```
np.random.seed(12)

races =   ["asian","black","hispanic","other","white"]

# Generate random data
voter_race = np.random.choice(a= races,
                              p = [0.05, 0.15 ,0.25, 0.05, 0.5],
                              size=1000)

voter_age = stats.poisson.rvs(loc=18,
                              mu=30,
                              size=1000)
```

```
# Group age data by race
voter_frame = pd.DataFrame({"race":voter_race,"age":voter_age})
groups = voter_frame.groupby("race").groups

# Etract individual groups
asian = voter_age[groups["asian"]]
black = voter_age[groups["black"]]
hispanic = voter_age[groups["hispanic"]]
other = voter_age[groups["other"]]
white = voter_age[groups["white"]]

# Perform the ANOVA
stats.f_oneway(asian, black, hispanic, other, white)
```

```
F_onewayResult(statistic=np.float64(1.7744689357329695),
pvalue=np.float64(0.13173183201930463))
```

```
import statsmodels.api as sm
from statsmodels.formula.api import ols

model = ols('age ~ race',                 # Model formula
            data = voter_frame).fit()

anova_result = sm.stats.anova_lm(model, typ=2)
print (anova_result)
```

```
             sum_sq     df         F    PR(>F)
race        199.369    4.0  1.774469  0.131732
Residual  27948.102  995.0       NaN       NaN
```

```
np.random.seed(12)

# Generate random data
voter_race = np.random.choice(a= races,
                              p = [0.05, 0.15 ,0.25, 0.05, 0.5],
                              size=1000)

# Use a different distribution for white ages
white_ages = stats.poisson.rvs(loc=18,
                              mu=32,
                              size=1000)

voter_age = stats.poisson.rvs(loc=18,
                              mu=30,
                              size=1000)
voter_age = np.where(voter_race=="white", white_ages, voter_age)

# Group age data by race
voter_frame = pd.DataFrame({"race":voter_race,"age":voter_age})
groups = voter_frame.groupby("race").groups

# Extract individual groups
asian = voter_age[groups["asian"]]
```

```
black = voter_age[groups["black"]]
hispanic = voter_age[groups["hispanic"]]
other = voter_age[groups["other"]]
white = voter_age[groups["white"]]
# Perform the ANOVA
stats.f_oneway(asian, black, hispanic, other, white)
```

```
F_onewayResult(statistic=np.float64(10.164699828386366),
pvalue=np.float64(4.5613242113994585e-08))
```

```
# Alternate method
model = ols('age ~ race',                 # Model formula
            data = voter_frame).fit()

anova_result = sm.stats.anova_lm(model, typ=2)
print (anova_result)
```

```
                sum_qq     df         F        PR(>F)
race       1284.123213    4.0  10.1647  4.561324e-08
Residual  31424.995787  995.0      NaN           NaN
```

The test result suggests the groups don't have the same sample means in this case, since the p-value is significant at a 99% confidence level. We know that it is the white voters who differ because we set it up that way in the code, but when testing real data, you may not know which group(s) caused the test to throw a positive result. To check which groups differ after getting a positive ANOVA result, you can perform a follow up test or "post-hoc test".

One post-hoc test is to perform a separate t-test for each pair of groups. You can perform a t-test between all pairs using by running each pair through the stats.ttest_ind() we covered in the lesson on t-tests:

```
# Get all race pairs
race_pairs = []

for race1 in range(4):
    for race2  in range(race1+1,5):
        race_pairs.append((races[race1], races[race2]))

# Conduct t-test on each pair
for race1, race2 in race_pairs:
    print(race1, race2)
    print(stats.ttest_ind(voter_age[groups[race1]],
                          voter_age[groups[race2]]))
```

```
asian black
TtestResult(statistic=np.float64(0.8386446909747979), pvalue=np.float64(0.402
asian hispanic
TtestResult(statistic=np.float64(-0.42594691924932293), pvalue=np.float64(0.6
asian other
TtestResult(statistic=np.float64(0.9795284739636), pvalue=np.float64(0.329887
asian white
```

```
TtestResult(statistic=np.float64(-2.318108811252288), pvalue=np.float64(0.020
black hispanic
TtestResult(statistic=np.float64(-1.9527839210712925), pvalue=np.float64(0.05
black other
TtestResult(statistic=np.float64(0.28025754367057176), pvalue=np.float64(0.77
black white
TtestResult(statistic=np.float64(-5.379303881281834), pvalue=np.float64(1.039
hispanic other
TtestResult(statistic=np.float64(1.5853626170340225), pvalue=np.float64(0.113
hispanic white
TtestResult(statistic=np.float64(-3.5160312714115376), pvalue=np.float64(0.00
other white
TtestResult(statistic=np.float64(-3.763809322077872), pvalue=np.float64(0.000
```

The p-values for each pairwise t-test suggest mean of white voters is likely different from the other groups, since the p-values for each t-test involving the white group is below 0.05. Using unadjusted pairwise t-tests can overestimate significance, however, because the more comparisons you make, the more likely you are to come across an unlikely result due to chance. We can adjust for this multiple comparison problem by dividing the statistical significance level by the number of comparisons made. In this case, if we were looking for a significance level of 5%, we'd be looking for p-values of 0.05/10 = 0.005 or less. This simple adjustment for multiple comparisons is known as the Bonferroni correction.

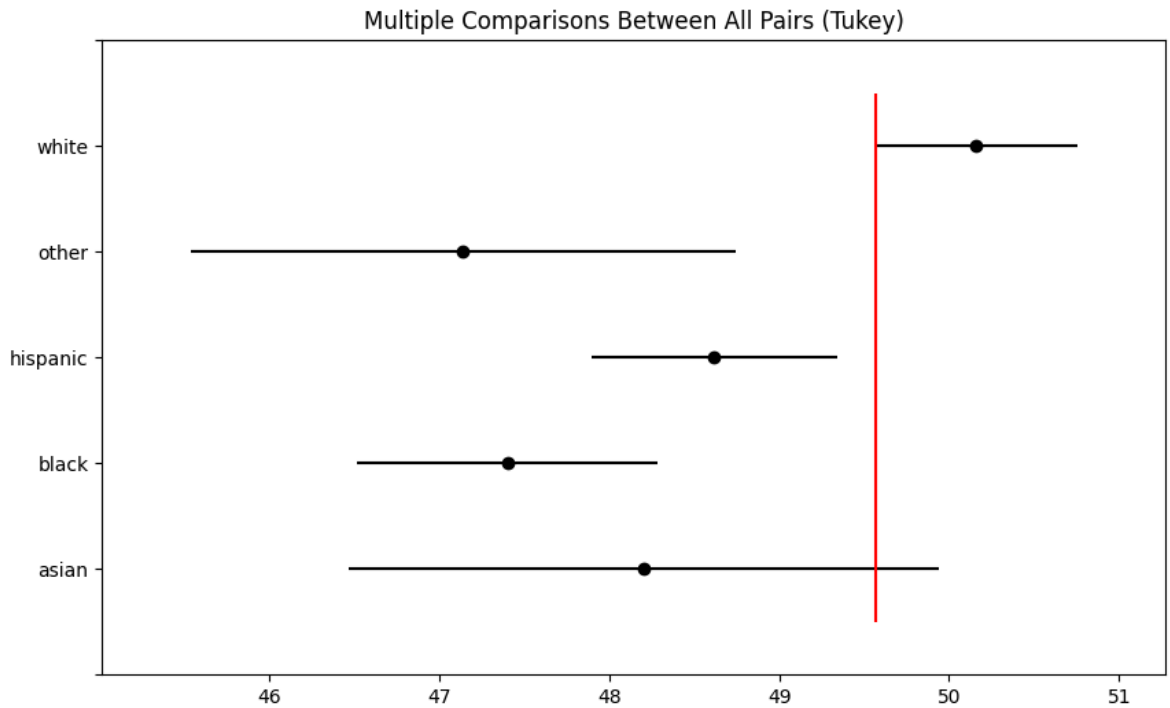
The Bonferroni correction is a conservative approach to account for the multiple comparisons problem that may end up rejecting results that are actually significant. Another common post hoc-test is Tukey's test. You can carry out Tukey's test using the pairwise_tukeyhsd() function in the statsmodels.stats.multicomp library:

Start coding or generate with AI.

```
from statsmodels.stats.multicomp import pairwise_tukeyhsd

tukey = pairwise_tukeyhsd(endog=voter_age,     # Data
                          groups=voter_race,   # Groups
                          alpha=0.05)          # Significance level

tukey.plot_simultaneous()    # Plot group confidence intervals
plt.vlines(x=49.57,ymin=-0.5,ymax=4.5, color="red")

tukey.summary()              # See test summary
```

Multiple Comparison of Means - Tukey HSD, FWER=0.05

group1	group2	meandiff	p-adj	lower	upper	reject
asian	black	-0.8032	0.9208	-3.4423	1.836	False
asian	hispanic	0.4143	0.9915	-2.1011	2.9297	False
asian	other	-1.0645	0.8906	-4.2391	2.11	False
asian	white	1.9547	0.1751	-0.4575	4.3668	False
black	hispanic	1.2175	0.2318	-0.386	2.821	False
black	other	-0.2614	0.9986	-2.7757	2.253	False
black	white	2.7579	0.0	1.3217	4.194	True
hispanic	other	-1.4789	0.4374	-3.863	0.9053	False
hispanic	white	1.5404	0.004	0.3468	2.734	True
other	white	3.0192	0.0028	0.7443	5.2941	True



The output of the Tukey test shows the average difference, a confidence interval as well as whether you should reject the null hypothesis for each pair of groups at the given significance level. In this case, the test suggests we reject the null hypothesis for 3 pairs, with each pair including the "white" category. This suggests the white group is likely different from the others. The 95% confidence interval plot reinforces the results visually: only 1 other group's confidence interval overlaps the white group's confidence interval.

1.Explain input and output characteristics for f_oneway function from stats library

Input Characteristics: This function takes an arbitrary number of array-like objects as positional arguments. Each argument represents a sample from a different group or category. For example, f_oneway(group1_data, group2_data, group3_data, ...). The data within each array should be numeric. Output Characteristics: It returns an object of type F_onewayResult which has two main attributes: statistic: This is the calculated F-statistic from the ANOVA. The F-statistic is a ratio of the variance between the group means to the variance within the groups. pvalue: This is the p-value associated with the F-statistic. It indicates the probability of observing an F-statistic as extreme as, or more extreme than, the one calculated, assuming the null hypothesis (that all group means are equal) is true.

2.Explain results of sm.stats.anova_lm fucntion.

Results of sm.stats.anova_lm function: When you use sm.stats.anova_lm from statsmodels on a fitted linear model (like an OLS model), it generates an ANOVA table, typically presented as a pandas DataFrame. Here's a breakdown of the common columns:

sum_sq (Sum of Squares): This column shows the variation attributable to each factor (e.g., 'race' in your example) and the residual (unexplained) variation. For factors: Represents the sum of squares due to the variation between the group means. For Residual: Represents the sum of squares due to variation within the groups (error). df (Degrees of Freedom): The number of independent pieces of information used to calculate the sum of squares for each factor. For a factor with k levels, df is k-1. For Residuals, df is N-k, where N is the total number of observations and k is the number of groups. F (F-statistic): This is the F-statistic for each factor, calculated as the mean square of the factor divided by the mean square of the residuals. It indicates how much the variance explained by the model (or a specific factor) is greater than the unexplained variance. PR(>F) (p-value): This is the p-value associated with the F-statistic for each factor. It tells you the probability of observing such an F-statistic if the null hypothesis (that the factor has no effect on the dependent variable) were true. A low p-value (typically < 0.05) suggests that the factor has a statistically significant effect.

3.What role ANOVA plays in data analytics?

Role of ANOVA in Data Analytics: ANOVA (Analysis of Variance) plays a crucial role in data analytics, particularly in inferential statistics, by:

Comparing Multiple Group Means: Its primary function is to determine if there are any statistically significant differences between the means of three or more independent groups. This is a common scenario in A/B testing, clinical trials, and comparing

demographic groups. Avoiding Type I Errors: If you were to conduct multiple pairwise t-tests to compare all combinations of groups, you would increase the likelihood of committing a Type I error (false positive). ANOVA performs a single, omnibus test, controlling the overall error rate. Identifying Factor Effects: It helps analysts understand if a categorical independent variable (or factor) has a significant impact on a continuous dependent variable. For example, does different training methods affect employee performance, or does different fertilizer types affect crop yield? Foundation for Post-Hoc Tests: When an ANOVA returns a significant p-value, it indicates that at least one group mean is different from the others, but it doesn't specify which groups differ. ANOVA results often lead to follow-up (post-hoc) tests (like Tukey's HSD) to pinpoint the specific differences between group pairs. Hypothesis Testing: It provides a robust framework for hypothesis testing, allowing data analysts to make statistically sound conclusions about group differences based on sample data. In essence, ANOVA is a powerful tool for exploring relationships between categorical and continuous variables, guiding further investigation, and making informed decisions based on data.

4.When to use one way ANOVA? Explain with real world case.

When to use One-Way ANOVA? YouOne-Way ANOVA is used when you want to compare the means of three or more independent groups based on one single categorical independent variable (factor). The goal is to determine if there's a statistically significant difference between the means of these groups on a continuous dependent variable.

Real-world case example for One-Way ANOVA:

Imagine a pharmaceutical company wants to test the effectiveness of three different dosages of a new drug (e.g., 10mg, 20mg, 30mg) on lowering blood pressure. They recruit 30 patients and randomly assign 10 patients to each dosage group. After a month, they measure the reduction in blood pressure for each patient.

Dependent Variable: Reduction in blood pressure (continuous). Independent Variable (Factor): Drug dosage (categorical, with three levels: 10mg, 20mg, 30mg). Here, a one-way ANOVA would be appropriate to determine if there is a significant difference in the average blood pressure reduction among the three different drug dosage groups. If the ANOVA is significant, it suggests that at least one dosage level has a different effect on blood pressure reduction than the others.

5.When to use 2 way ANOVA ? Explain with real world case.

When to use Two-Way ANOVA? Two-Way ANOVA is used when you want to examine the effect of two categorical independent variables (factors) on a continuous dependent variable. It not only assesses the main effect of each independent variable

but also investigates whether there's an interaction effect between the two independent variables.

Real-world case example for Two-Way ANOVA:

Let's say a marketing team wants to study the effectiveness of a new advertisement campaign on product sales. They hypothesize that both the type of advertisement (e.g., TV commercial, social media ad) and the region where it's shown (e.g., urban, rural) might influence sales, and they also suspect that the combination of ad type and region might have a unique impact.

They run an experiment where different combinations of ad types and regions are tested, and sales data is collected.

Dependent Variable: Product sales (continuous). Independent Variable 1 (Factor 1): Type of Advertisement (categorical, with two levels: TV commercial, Social media ad). Independent Variable 2 (Factor 2): Region (categorical, with two levels: Urban, Rural). A two-way ANOVA would be suitable here to:

Determine if there's a significant main effect of Ad Type on sales (i.e., does one ad type generally lead to higher sales than the other, regardless of region?). Determine if there's a significant main effect of Region on sales (i.e., do sales generally differ between urban and rural areas, regardless of ad type?). Determine if there's a significant interaction effect between Ad Type and Region (i.e., does the effect of an ad type on sales depend on the region it's shown in? For example, a TV ad might be more effective in rural areas, while a social media ad might be more effective in urban areas, or vice-versa). In summary, choose one-way ANOVA for comparing groups based on one factor, and two-way ANOVA for comparing groups based on two factors, especially when you're interested in their combined (interaction) effect.